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Out-of-Distribution Material Property Prediction Using Adversarial Learning

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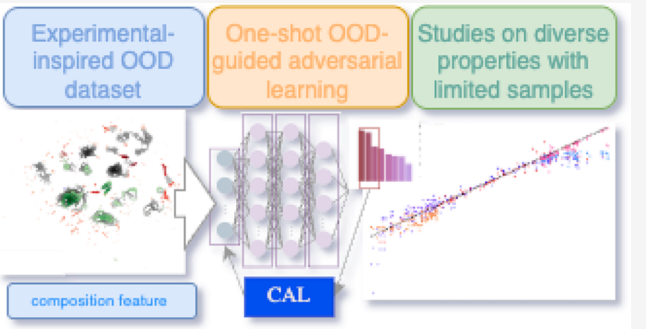
ABSTRACT: The accurate prediction of material properties is crucial in a wide range of scientific and engineering disciplines. Machine learning (ML) has advanced the state of the art in this field, enabling scientists to discover novel materials and design materials with specific desired properties. However, one major challenge that persists in material property prediction is the generalization of models to out-of-distribution (OOD) samples, i.e., samples that differ significantly from those encountered during training. In real-world materials discovery, OOD scenarios often arise when applying ML to predict additional materials within a newly explored region originating from a few experimental samples. In this paper, we explore the application of advancements in OOD learning approaches to enhance the robustness and reliability of material property prediction models. We propose and apply the Crystal Adversarial Learning (CAL) algorithm for OOD materials property prediction, which generates synthetic data during training to guide learning toward those samples with high prediction uncertainty. We further propose an adversarial learning-based targeted approach to make the model adapt to a particular OOD data set, as an alternative to traditional fine-tuning. Our experiments suggest that our CAL algorithm can be effective in ML scenarios with limited samples, which commonly occur in materials science. Our work provides an important step toward improved OOD learning and materials property prediction and highlights areas that require further exploration and refinement.



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■ INTRODUCTION

The accurate characterization and prediction of material properties are pivotal for various applications, ranging from renewable energy systems and drug design to electronic devices and aerospace engineering. Traditionally, material property determination has relied on costly, time-consuming experimental methods. However, with the advent of machine learning, particularly deep learning, it has become feasible to develop predictive models that can efficiently analyze vast amounts of material data and provide accurate property predictions. For over a century, materials scientists have accumulated a large number of crystal structures through laborious experimentation, leading to modern crystal structure databases such as the Inorganic Crystal Structure Database (ICSD),[1−3] containing over 200,000 structures identified since 1913 with about 7000 structures added each year, and more open databases such as the Materials Project,[4] containing over 120,000 inorganic materials along with their properties. Despite the successes achieved by machine learning models for material property prediction,[5−9] these models often struggle to generalize from their training data to test data that may be out of distribution (OOD)−drawn from areas of the feature space dissimilar to the training data. These OOD predictions can be markedly inaccurate, leading to unreliable results and potentially hindering the development of novel materials.[10] Addressing this challenge requires adopting methodologies that have proven effective in other areas.

Adversarial learning has been widely applied in various domains, including soft sensor development, to address challenges in data scarcity and modeling accuracy. In ref 11, the authors proposed a diffusion-based adversarial learning framework that leverages LSTM and one-dimensional (1D) convolutional layers to capture spatiotemporal dependencies, enabling the generation of high-quality time-series samples to augment limited data sets. In ref 12, an adversarial active learning method was introduced, combining a latent-enhanced variational autoencoder (LEVAE) and a discriminator to identify and incorporate informative unlabeled samples, thereby improving predictive performance with minimal labeling costs. Furthermore, in ref 13, authors employed a hierarchical transfer learning algorithm with an adversarial network consisting of a scaled conjugate gradient-based

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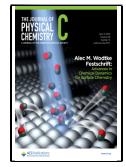

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generator and a cross-entropy error-based discriminator, alongside a Granger causality analyzer, to refine data quality and provide interpretable insights into input-output relationships. These studies demonstrate the versatility and effectiveness of adversarial learning in addressing modeling challenges across domains. While Generative Adversarial Networks (GANs) have been applied in materials science,[14−16] their use has predominantly focused on material generation rather than addressing out-of-distribution (OOD) property prediction challenges. Building on the demonstrated versatility of adversarial learning in other domains, we explore its potential for materials science and ability to tackle the unique challenges posed by OOD data in materials discovery. The adaptation of these techniques to materials science requires a tailored approach that considers the intricate relationships between material composition, structure, and properties. This paper explores such an adaptation through the development of the Crystal Adversarial Learning (CAL) framework, designed to enhance the robustness of material property predictions in OOD settings.

Current state-of-the-art machine learning models in materials science operate under the classical paradigm, in which data is assumed to be independently, identically distributed (i.i.d.) under a single data distribution. However, materials design problems often violate this core assumption. For example, material property optimization−finding materials that maximize or minimize a property−almost by definition involves predictions on new inputs that are not typical of the training data set. Although there is some research conducted on molecular out-of-distribution (MOOD) problems,[17−19] there are only a few studies focusing on the crystal material OOD property prediction problem,[10,20] in which the same challenge arises: training and testing data differ greatly in real-world deployment.[17,19] Existing ML models usually have poor performance for such OOD inputs.[21−23] Direct optimization using a trained ML surrogate, even for standard properties like formation energy, often leads to exploiting inaccuracies in the model instead of correctly discovering new stable materials.[24] Even with techniques such as fine-tuning, traditional ML models will often either perform poorly or overfit toward the target data set. The quality of the OOD data set also plays a vital role in enhancing the performance of the model, as Zou et al. discovered in their OOD research:[25] an OOD test set with an inappropriate difficulty level causes poor performance. To address this issue, they used the adaptive calibrator ensemble (ACE) algorithm which combines the outputs of two calibrators trained on data sets containing samples with high prediction uncertainty or rare feature distributions. In our work we adopt the partial sampling algorithm[26] to actively adjust the proportion of high-uncertainty or challenging samples included in our training set. Another related work is Qi et al.[24] which focused on training conservative objective energy models for gradient-based optimization for material structure prediction. Although we share the similar idea of utilizing gradients, their work is more toward exploring the search space with a latent space conservative model aimed to obtain the optimized crystal structure. Other related works in the MOOD field including Lee et al.[18] are generative models that seek new materials in a targeted OOD region.

In machine learning, there are three main types of out-of-distribution (OOD) problems: (1) Blind OOD prediction, where no data from the target domain is available; (2) OOD with limited unlabeled data from target domain; and (3) OOD with limited labeled data from target domain.[27,28] Recent efforts have focused on addressing blind OOD prediction in molecular studies,[17−19] while other research in materials science has explored OOD detection, model improvement,[29] and bias mitigation strategies.[30] In real-world materials discovery, the OOD scenario typically aligns with case 3, where new material families are initially identified through a limited number of experimental samples. Machine learning models are then applied to predict additional materials within this emerging domain. Our model is specifically designed to handle this practical OOD scenario by employing adversarial learning with a small set of available samples, significantly improving performance in discovering new materials. While our method is not optimized for blind OOD prediction, this scenario is uncommon in material research. Instead, our approach addresses a prevalent challenge in the field: the availability of limited but critical OOD data, enabling the model to explore and predict additional materials in under-explored regions.

This OOD material property prediction problem needs specifically designed OOD data sets for evaluation. Here we introduce such OOD data sets for our experiments, curated from a larger training set. We define the 145 composition-based physical attributes as the input X and the target property formation energy as y for all samples extracted from the Materials Project data set.[4] Our crafted OOD data sets can be categorized into three different classes: *Covariate Shift* (X shift), *Prior Shift* (Y shift), and *Relation Shift* ($P(Y|X)$ shift).[31] OOD samples pose great challenges for current ML models based on i.i.d. data. Table 1 shows the high variation in the performance of several state-of-the-art composition-based neural network models for material property prediction, emphasizing the pressing need to address OOD prediction challenges.

Table 1. Performance of State-of-the-Art Models on Our Out-of-Distribution (OOD) Test Sets for Formation Energy Prediction (Units: eV), Measured in Mean Absolute Error (MAE)

data set[a]	CrabNet[b]	MODNet	IRNet-48
training set	0.0412	0.066	0.002
test set (i.i.d)	0.0624	0.125	0.411
covariate shift	0.260	0.2659	0.408
prior shift	0.953	0.9089	2.091
relation shift	0.571	DNF[c]	1.188

[a]The data set x attribute is not normalized in this table. [b]CrabNet generates all features internally from the formula. [c]MODNet performs feature selection, and the Piezoelectric materials are missing some features as NaN. The feature selector removes columns that cause misalignment between the data set dimension and the network dimension.

In this work, we propose the crystal adversarial learning (CAL) pipeline to address the OOD material property prediction problem based on the stable adversarial learning algorithm (SAL) proposed in ref 32. The SAL approach constructs an uncertainty set by first assessing feature stability, how reliably they correlate with the target. It then generates adversarial samples by perturbing unstable features using network gradients and optimizes seed sample weights through gradient descent. Our CAL algorithm effectively identifies high correlation features that can be used to screen challenging

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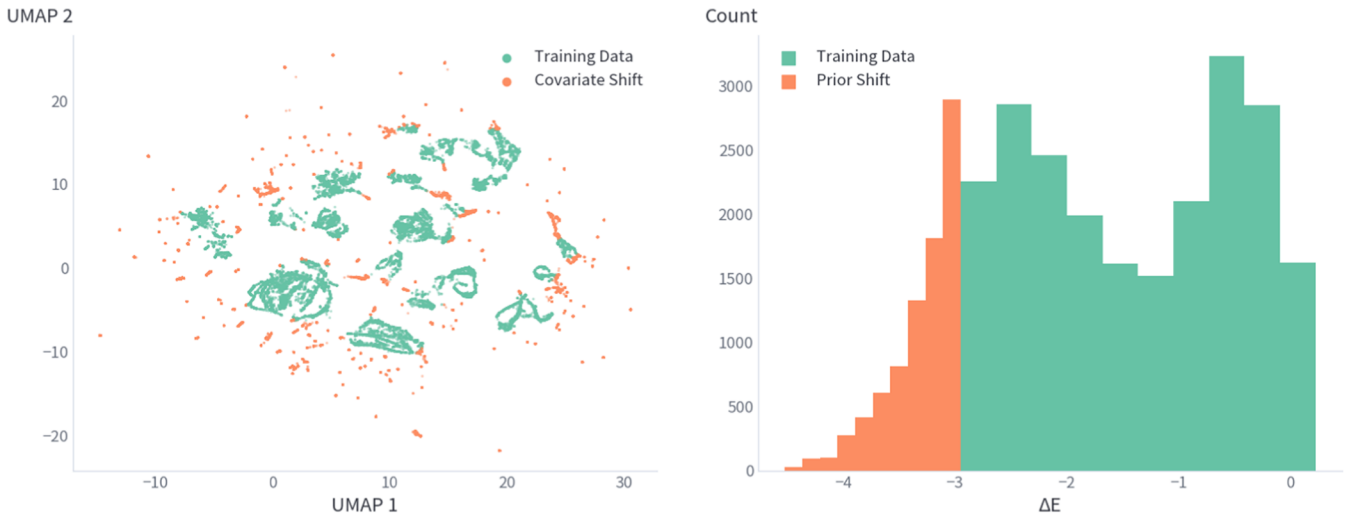

(a) Covariate shift samples (b) Prior shift dataset (for formation energy)

Figure 1. Data set selections for covariate and prior shifts. The covariate shift samples are selected from the periphery of the low-dimensional surface formed by the data. The prior shift data set consists of outliers in formation energy.

OOD samples from the seed training samples for adversarial learning, expecting the model to achieve better performance for future real-world OOD samples. We extend this approach through partial sampling, iteratively selecting samples from the training set with high training losses and training specifically on those difficult samples. The combination of these two methods improves our CAL model's OOD performance. CAL is also able to target specific OOD data sets and generate adversarial samples based on them. The CAL method is designed to operate independently, achieving its performance without the necessity of prior training on a different data set. This unique aspect of the CAL approach enables it to directly adapt to the specifics of the material data sets at hand, allowing for effective learning from the available data without relying on external pretrained models. This method offers an alternative to the fine-tuning technique used in conventional machine learning (ML), as CAL training utilizes a one-shot process. This targeted approach allows us to investigate how all three types of distribution shifts affect the model performance. By understanding these relationships, we aim to develop more robust and generalizable predictive models for material discovery, especially for special materials such as piezoelectric crystals and special properties such as phonon peaks.

Our work makes three significant contributions to the field of material property prediction. First, we identify the urgent need to address the limitations of existing machine learning models when dealing with out-of-distribution (OOD) data. Second, we propose CAL, an innovative approach that integrates adversarial learning with training set ranking, which enhances model robustness under various test scenarios characterized by different input and target distributions. Finally, CAL demonstrates potential in handling low-data problems and distribution shifts, showing improvements over competitive baselines in specific scenarios.

■ METHODS

OOD Data Sets. We focus on the OOD material property prediction problem and use the formation energy and bandgap for our main experiments. Using Matminer,[33] we select 84,190 materials from the Materials Project database.[4] For each sample, we generate 145 composition-based physical attributes.[34] We call the input information X and the regression target y. X is a set of statistics describing a material's composition, and y is a property of that material—formation energy or bandgap. There are three kinds of potential distribution shifts: *Covariate shift*, *Prior shift*, and *Relation shift*.[31] Covariate shift is defined as a change in the distribution of the input data X between training and inference. Prior shift, also known as *label shift*, occurs when the distribution of y shifts from training to inference without a corresponding change in X. Relation shift, known by many other names including *concept shift*, denotes a change in the joint distribution of X and y—how the features X relate to the target y without a shift in the individual marginal distributions of X or y. In practice, real-world problems exhibit all three shifts to some degree, and these categories are not absolute. In our previous work,[35] we used the global mapping to show the relationship between X and y. We consider the distribution of outputs y as conditional on the inputs X. In this notation, covariate shift can be thought of as shift in $P(X)$, prior shift as shift in $P(y)$, and relation shift as shift in $P(y|X)$.

We constructed several data sets to exhibit these distinct OOD challenges. For covariate shift, we use a low-dimensional representation of the data with UMAP,[36] a dimension reduction technique,and select data points located on the periphery of the clusters in the two-dimensional (2D) UMAP representation, as these peripheral points indicate covariate shift. (Figure 1a). For prior shift, we select the extreme values of y (Figure 1b). For the relation shift set, we introduce a data set for piezoelectric materials. Piezoelectric materials serve as a compelling case study because of their rarity and exceptional chemistry. They have unique relationship between the magpie feature and their formation energy and bandgap. The currently available data, which solely originates from experiments, is not readily amenable to traditional machine learning techniques due to its inherent variability and potential noise. That same uncertainty, however, makes machine learning models even more important for research and further exploration.

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To account for variations in composition features, we initially normalize our data. Then, as mentioned above, the OOD data sets can be categorized into three different classes: covariate shift (X shift), prior probability shift (Y shift), and relation shift ($P(Y|X)$ shift).

For the prior shift data set, the existing data set is first sorted by the property values y. Rather than employing a random split as is usually done in conventional ML studies, we partition the data set based on a defined percentage, such as the top 20% of y values. This procedure results in a prior shift data set where the target (y) of the test set remains unseen by the network. Figure 1(b) illustrates the generation of this supervised prior shift set for the y values of the formation energy.

Constructing our covariate shift data set is more involved because our input is 145-dimensional and cannot be sorted. Instead, we use UMAP[36] to reduce our data to 2 dimensions and then apply a kernel density estimate to select the points farthest from the main data distribution based on a defined percentage. The results and the embedding are shown in Figure 1(a), where the orange data points represent the selected out-of-distribution (OOD) samples.

To investigate relation shift ($P(y|X)$ shift), we opted for piezoelectric materials. This choice is motivated by two key factors. First, piezoelectric materials exhibit distinctive patterns in the Magpie feature space due to their dependence on polarization and crystal symmetry. These patterns differ significantly from those of nonpiezoelectric materials, which dominate the our data set, making piezoelectric materials a challenging choice for OOD evaluation. This distinct characteristic allows us to effectively explore how changes in this relationship impact model performance. Second, the availability of piezoelectric material samples is limited, reflecting a common bottleneck encountered in real-world materials science research. This limited data scenario provides a valuable opportunity to test our model's capability in handling situations with scarce samples.

After removing all the OOD samples mentioned above, we constructed the conventional ML training and testing set using an 85:15 ratio to ensure that all test sets have roughly the same size. Finally, 10% of the training set was used for validation. There are four main test sets for evaluation. All test sets were unseen during training. They are listed below, and their distribution is shown in Figure 2.

- Independently and identically distributed (i.i.d.) test set
- Covariate shift test set
- Prior shift test set
- Relation shift (piezoelectric) test set

In addition, we also tested our models on special data sets with limited data size. We selected the last phonon peak subset with 1265 samples, the refractive index subset with 4764 samples, and the shear stress subset with 10,987 samples for our experiments. Different guidelines were used to determine the OOD prior shift sets for these smaller data sets. Specifically, we selected the top 10% and bottom 10% of target values as our prior shift sets. Since these data sets are already subsets of materials with specific properties, there is no clear subset to select for relation shift, and thus, no relation shift sets were chosen.

Crystal Adversarial Learning (CAL). Our CAL algorithm adapts the Stable Adversarial Learning (SAL) pipeline[32] and, in this work, is applied with individual residual networks for material property prediction. The core concept involves creating an uncertainty set by distinguishing features based on the stability of their correlations with the target. Adversarial samples are generated by perturbing unstable features using network gradients, and seed sample weights are optimized through iterative gradient descent. Furthermore, we employ partial sampling techniques to iteratively select out-of-distribution (OOD) samples with high training losses. During this process, guided adversarial samples are generated, and partial sampling is used to enhance our model's ability to handle various OOD challenges.

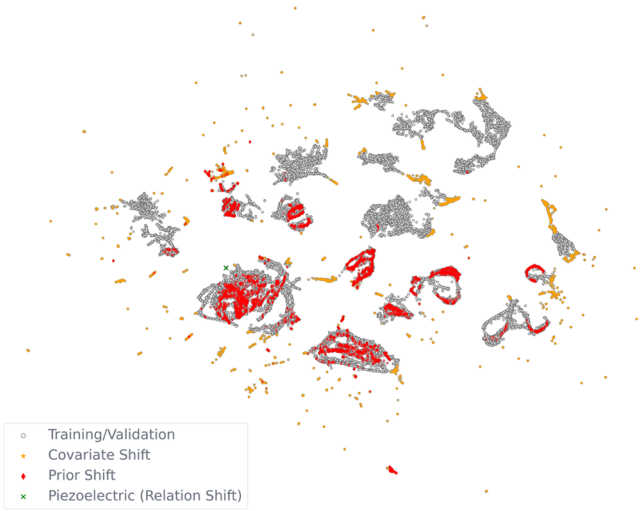

Figure 2. UMAP embedding of the data, illustrating variance in both X and y (formation energy). The covariate shift samples (orange) are distributed across the periphery, showing some overlap with the majority of the samples. The prior shift samples (red) form their own distinct clusters, while the piezoelectric (relation shift) samples (green) cluster tightly together remaining distinct from the training/validation set, indicating a distinct relationship influenced by unique material properties.

Figure 3 depicts the CAL pipeline and experimental setup. We first initialize the network with training samples for several epochs. Subsequently, samples are sorted based on their loss at every ith epoch, initiating at the ith epoch. Concurrently, the CAL module extracts gradients as partial samples traverse the network. At every jth epoch, the CAL module generates adversarial samples for the network. The CAL framework employs two types of loss: (1) observed sample loss, which evaluates the models performance on the original training samples, and (2) generated sample loss, which measures the models ability to predict adversarially generated samples that simulate OOD scenarios. Together, these losses enable the model to balance learning from in-distribution data and preparing for OOD challenges. The training process continues until the average loss of i.i.d. validation sets ceases to improve for k consecutive epochs. This criterion is implemented to ensure the stability of the training distribution samples and marks the conclusion of the training phase. Another scenario is when working on a small data set, such as piezoelectric materials. The limited samples are a huge challenge for current ML prediction models. Fine-tuning with such a small data set brings issues, such as overfitting and loss of robustness. Our CAL pipeline is able to target those data sets and generate adversarial samples based on them. In this setup, all adversarial samples are generated using one batch of piezoelectric samples, while others are not disrupted. In this way, the model has

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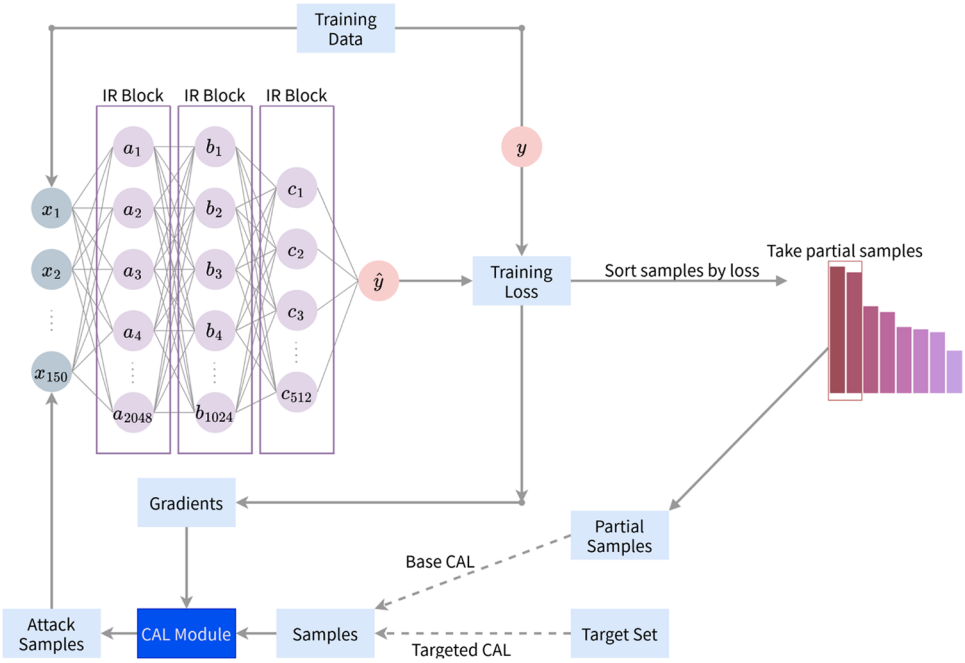

Figure 3. Schematic diagram of the Crystal Adversarial Learning (CAL) framework for out-of-distribution (OOD) materials property prediction. The process begins with training data passing through a neural network to generate predictions ($\hat{y}$), which are compared against true labels (y) to compute the training loss. Samples are then sorted based on loss values, and a subset of high-loss samples is selected for adversarial augmentation using the CAL module. Attack samples are perturbed using gradient-based methods, generating new adversarial samples that challenge the model. The generated samples are integrated into the training process through both Base CAL (standard adversarial augmentation) and Targeted CAL (focused adversarial selection). This iterative process enhances model robustness against distributional shifts in materials data.

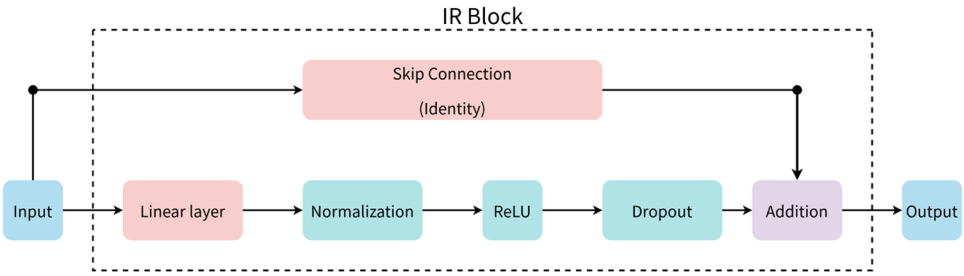

Figure 4. Schematic diagram of the IR-net block architecture. A skip connection is added from the input to the output, enabling residual learning. If the input and output dimensions differ, a simple linear layer is applied to project the input to the appropriate dimension before the skip connection. The main network consists of a linear layer, normalization, ReLU activation, dropout, and an addition step that merges the transformed input with the skip connection, enhancing stability and gradient flow during training.

enough samples to learn from while avoiding overfitting. We take an exponential moving average (EMA) of epoch checkpoints for our final model.[37] EMA is widely used for better generalization in normal training with SGD; we find that, for CAL training, EMA is even more useful. We hypothesize that CAL, with partial sampling rotating inputs periodically, induces training to go between different local minima representing different parts of the data. Averaging these weights at the end provides more robust performance than any individual model.

Targeting-Based Fine-Tuning for OOD SAL Algorithm. One of the major improvements in our CAL algorithm compared to SAL is the introduction of a targeting mechanism for fine-tuning a pretrained model to adapt to an OOD test set. In standard fine-tuning, a small batch of samples from the test set domain is typically used to fine-tune the model. However, this approach can easily lead to overfitting due to the limited number of fine-tuning samples. To address this, we propose a targeting-based fine-tuning mechanism, which uses selected test domain samples to guide the adversarial sample generation in SAL. This approach ensures that the generated adversarial samples are more similar to the test domain samples. The selected target domain samples serve as the seed samples for adversarial sample generation, in contrast to the original SAL, which uses all training samples as seeds. This modification not only makes our CAL algorithm more scalable but also guides the weighting toward target domain samples, significantly improving OOD performance compared to standard fine-tuning.

Partial Sampling. We adapted the partial sampling approach from Magar et al.[26] Periodically during training, the training set is sorted based on the loss computed by the current model. The batches with the highest loss, referred to as partial samples, are the only training batches used until new partial samples are regenerated. This entire process is depicted in Figure 3.

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J. Phys. Chem. C 2025, 129, 6372−6385

Layer Block in Our CAL Neural Network Model. We adopted and modified IR-net[9] as the backbone for our model, as shown in Figure 3. The original IR-net achieved state-of-the-art results with an intricate architecture comprising 48 layers of individual residual blocks. However, for our data, we found that this depth was counterproductive, so we opted for a more compact architecture.

In our refinement, we streamlined the network to three layers with individual residual blocks, having dimensions of 2048, 1024, and 512, respectively. The structure of the individual residual blocks is shown in Figure 4. Unlike conventional residual neural networks, we add the original input to the output after each block. The input vector is first processed through a fully connected layer, followed by layer normalization (we use LayerNorm[38] instead of the original batch normalization) and ReLU activation. Additionally, we found that dropout is essential for mitigating overfitting in our data set. Our experiments demonstrate that our improved IR-net, which is wider but shallower, achieves similar or even better prediction performance compared to the original IR-net, with improved computational efficiency.

Adversarial Attack Algorithm for OOD Machine Learning (AA). We assimilated the SAL algorithm from Liu et al.[32] for OOD materials property prediction. The SAL algorithm constructs an uncertainty set by categorizing covariates into stable and unstable groups based on their correlation with the target variable. It generates adversarial samples by perturbing the unstable covariates (features) using network gradients and optimizes covariate weights through iterative gradient descent to balance both average and worst-case performance across training environments. It is important to note that the algorithm relies on the availability of the training domain distribution to identify covariate feature vectors. The original algorithm also utilized Wasserstein Distributionally Robust Learning (WDRL), which aims to create an uncertainty set characterized by the stability of covariates, thereby inducing stronger adversarial perturbations on unstable covariates. In their work, the algorithm was tested on a real-world house price prediction problem using simple, fully connected single-layer perceptrons. However, certain challenges remain unresolved. Due to the nature of this method, the original adversarial data generation was not optimized for computational complexity. As a result, it has limited scalability, and its performance on large, complex data sets remains unknown.

From the framework of *Wasserstein Distributionally Robust Learning* (WDRL),[39] the original SAL architecture further characterizes the uncertainty set as anisotropic based on the stability of covariates across multiple environments. This induces stronger adversarial perturbations on unstable covariates than on stable ones. The WDRL framework is built on two key assumptions:

Assumption 1 suggests that there exists a decomposition of all the covariates, $X = \{S, V\}$, where S represents the stable covariate set and V represents the unstable one. The correlation between stable covariates S and the target Y remains invariant across environments.[40] This is supported by the success of deep learning models in material property prediction, where high accuracy rates indicate the presence of stable covariates that the network can learn.

Assumption 2 posits that there exists at least one distribution on the boundary whose marginal distribution on S differs from the center distribution of the original uncertainty set.[32] This assumption is satisfied by the artificially created distribution shift in our OOD problem.

If stable covariates are perturbed in an adversarial learning process, the target (y) should change proportionally to preserve the underlying relationship. In practice, however, we do not have access to the target values corresponding to the perturbed stable covariates. Consequently, optimizing under an isotropic uncertainty set that includes perturbations on both stable and unstable covariates typically reduces the model's confidence and produces meaningless results. This issue is addressed by assigning higher weights to stable covariates in the cost function, thereby constructing a more reasonable uncertainty set that avoids ineffective perturbations. The combined objective function below optimizes both the covariate differentiation process and the adversarial training of model parameters

$$\min_{\theta\in\Theta} \sup_{Q:W_{c_w}(Q,P_0)\le\rho} \mathbb{E}_{X,Y\sim Q}[l(\theta; X, Y)] \tag{1}$$

where

$$c_w(z_1, z_2) = \|w \odot (z_1 - z_2)\|_2^2 \tag{2}$$

and

$$w \in \arg\min_{w\in\mathcal{W}} \left\{ \frac{1}{|\mathcal{E}_{\mathrm{tr}}|} \sum_{e\in\mathcal{E}_{\mathrm{tr}}} \mathcal{L}^e(\theta) + \alpha \max_{e_p, e_q \in\mathcal{E}_{\mathrm{tr}}} \mathcal{L}^{e_p} - \mathcal{L}^{e_q} \right\} \tag{3}$$

P_0 denotes the training distribution, W_{c_w} denotes the Wasserstein distance with transportation cost function c_w defined as eq 3,

$$\mathcal{W} = \left\{ w\colon w \in [1, +\infty)^{m+1} \&\& \min(w^{(1)}, \cdots, w^{(m+1)}) = 1 \right\}$$

denotes the covariate weight space ($w^{(i)}$ denotes the ith element of w), and $\mathcal{L}^e$ denotes the average loss in environment $e \in \mathcal{E}_{\mathrm{tr}}$. α is a hyperparameter that trades off between average performance and the stability.[32]

The adversarial sampling objective function is

$$R(\theta(w)) = \frac{1}{|\mathcal{E}_{\mathrm{tr}}|} \sum_{e\in\mathcal{E}_{\mathrm{tr}}} \mathcal{L}^e(\theta(w)) + \alpha \max_{e_p, e_q \in\mathcal{E}_{\mathrm{tr}}} (\mathcal{L}^{e_p} - \mathcal{L}^{e_q}) \tag{4}$$

where α is the hyper-parameter. $R(\theta(w))$ contains two parts: the first is the average loss in multiple training environments; the second reflects the max margin among environments, which reflects the stability of $\theta(w)$, since it is easy to prove that $\max_{e_p,e_q\in\mathcal{E}_{\mathrm{tr}}} \mathcal{L}^{e_p}(\theta(w)) - \mathcal{L}^{e_q}(\theta(w)) = 0$ if and only if the errors among all training environments are same. Originally α is used to adjust the trade off between average performance and stability.

Technical Improvements for Increasing SAL Scalability. The original SAL algorithm[32] was designed for small models and data sets, allowing for the straightforward calculation of all necessary gradients. However, our data sets and models are significantly larger, necessitating several modifications to make SAL feasible. Instead of generating an adversarial copy of the entire data set, we subsample a portion for adversarial data generation. This approach is analogous to the standard use of stochastic gradient descent (SGD) rather than full gradient descent. Additionally, to avoid the memory-

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intensive task of materializing the full Jacobian, we compute only the necessary Jacobian-vector products.

The gradient used to optimize w, the SAL perturbation weights on features in the data set, is given as eq 11 in ref 32, where X_A is the adversarial data and θ represents the model parameters

$$\frac{\partial R(\theta(w))}{\partial w} = \frac{\partial R}{\partial \theta}\frac{\partial \theta}{\partial X_A}\frac{\partial X_A}{\partial w} \tag{5}$$

The original SAL implementation stores $\frac{\partial \theta}{\partial X_A}$ in memory before computing this product. For models with large parameter counts, this full Jacobian does not fit in memory as it contains $\dim(\theta) \times \dim(x)$ entries. Our implementation addresses this limitation by computing the vector-Jacobian product $\frac{\partial R}{\partial \theta}\frac{\partial \theta}{\partial X_A}$ directly, which has only $\dim(x)$ entries. This optimization enables the application of SAL to models with many thousands of parameters without encountering memory issues.

The loss in (eq 5) can be further approximated by using only a subset of the parameters θ. Although such approximations are not necessary for our work after the optimizations discussed above, we explore their effects in our ablation study by approximating the gradient using the linear weights of the first or last layer. Additionally, we find that normalizing the input X can enhance the scalability of SAL. All these technical improvements have enabled us to deploy SAL effectively in modern deep learning networks for material property prediction, as utilized in our CAL model.

In this study, we implemented smoothed L_1 loss, also known as Huber loss, due to its advantageous properties in machine learning applications. This loss function effectively combines the strengths of L_1 and L_2 losses, providing robustness against outliers while simultaneously maintaining gradient stability. For small errors, it behaves like L_2 loss, ensuring smoothness and differentiability near zero. For larger errors, it transitions to L_1 loss characteristics, thereby mitigating the impact of outliers. This dual nature not only prevents gradient explosion during training but also often results in accelerated and more stable convergence compared to pure L_1 loss. Given these attributes, smoothed L_1 loss is particularly effective for regression tasks, making it an optimal choice for our CAL pipeline training. Simultaneously, we used standard mean absolute error (MAE) for test set evaluation.

Experimental Design. Data Sources. The data sets utilized in this study were sourced from the Materials Project database,[4] containing 84,190 samples. Each sample was characterized by properties such as formation energy and bandgap, which were selected as the primary targets for evaluation. Additional subsets of materials, including those related to last phonon peak, shear modulus, and refractive index, were sourced from the Matbench leaderboard.[41] These subsets were specifically selected to test the model's performance on limited and challenging data sets.

Preprocessing Steps. For each sample, 145 composition-based physical attributes were extracted using the Matminer toolkit.[33] These features were normalized to ensure consistency across samples. To create out-of-distribution (OOD) data sets, the following strategies were applied:

Covariate Shift: Data points were selected from the periphery of the low-dimensional UMAP representation of the data set, identifying samples far from the training distribution. Prior Shift: Extreme values of the target property (e.g., top or bottom 10%) were isolated to form the test set. Relation Shift: Specialized data sets, such as piezoelectric materials, were used to evaluate shifts in the relationship between features and target properties. After constructing the OOD data sets, the remaining data was split into training (85%), validation (10%), and i.i.d. test (15%) sets to ensure rigorous evaluation.

Evaluation Metrics. The primary evaluation metric for model performance was mean absolute error (MAE), calculated separately for i.i.d. test sets and each OOD shift category (covariate, prior, and relation shifts). This approach enabled a comprehensive assessment of the model's generalization capabilities and its ability to handle diverse distributional scenarios. We also conducted multiple independent training runs with different random seed initializations, as presented in Supporting Table S4.

■ RESULTS

Standard ML Models Have Poor Performance on OOD Data Sets. Table 1 summarizes the performance of three deep learning models—CrabNet, MODNet, and IRNet-48[5,6,9]—across various test sets, highlighting their susceptibility to out-of-distribution (OOD) data. All models achieve low error rates (around 0.002−0.06 eV) when trained and validated on the training data set ("Training Set") and the i.i.d test set. However, their performance significantly degrades on our OOD data sets ("Covariate Shift", "Prior Shift", "Relation Shift"), with error rates increasing substantially across all models, ranging from 0.224 eV to as high as 2.091 eV.

The models' performances are acceptable over the random split and covariate shift data sets. The covariate shift does not pose a significant challenge for neural network models, as the shift in the X domain involves physical real-world data where the OOD samples are not far from the majority of the data distribution. In contrast, the prior shift selects OOD samples in the Y domain that are unfamiliar to the model, although their X features are still within the training distribution, as shown in Figure 2. The blue dots in the figure represent the prior shift samples, which overlap with the training samples shown as gray dots. The relation shift, marked in yellow in Figure 2, is characterized by limited samples. The piezoelectric materials selected for this shift are considered special within materials science studies, even though their X and Y are in-distribution. This is likely why the relation shift performance falls between that of the prior shift and covariate shift for the benchmark models.

Model-wise comparisons reveal that CrabNet and MODNet outperform the traditional neural network model IRNet-48. CrabNet, which uses attention-based element and atomic number representations, and MODNet, which employs hierarchical properties grouping, exhibit better performance than IRNet-48. IRNet-48 demonstrates a severe overfitting problem, with a training loss as low as 0.002 eV compared to a validation loss (random splitting) of 0.441 eV. This indicates that all three models struggle with generalizing to unseen data distributions, underscoring that our OOD data set presents a meaningful challenge for current models.

OOD Performance of the CAL Algorithm. Table 2 summarizes the OOD performance of the baseline models and fine-tuning models, comparing our targeted CAL method with the conventional IR-net model. When comparing the IR-net

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J. Phys. Chem. C 2025, 129, 6372−6385

Table 2. Out-of-Distribution (OOD) Performance (MAE) Comparison: Our CAL Method Maintains Stability during Training Compared to the Baseline for Predicting Formation Energy (units: eV)[a]

	train set	random shift	covariate shift	prior shift	relation shift
IR-net baseline[b]	0.0009	0.0863	0.2448	1.3672	0.2222
CAL iid	0.2111	0.5424	0.4925	1.3298	0.6115
IR-net fine-tune[c]		0.1657	0.2005	0.1471	0.1907
targeted CAL[c]		0.3387	0.1701	0.1216	0.0932
fine-tune % Improvement		−92.01%	18.09%	89.24%	14.17%
targeted CAL % Improvement		37.55%	65.46%	90.85%	84.75%

[a]For all targeted data sets, our CAL pipeline consistently outperforms traditional fine-tuning methods across various OOD shifts. Training was performed using smoothed L1 loss, and model performance was evaluated using mean absolute error (MAE). [b]This model serves as a baseline for fine-tuning. [c]Each column in these rows corresponds to different model targeted specific fine-tune/target sets. Training set does not apply due to the fine-tune will have loss of the OOD set while the targeted CAL's loss including the training set and the OOD set. the I.I.D. random split set will loss are the average of the 3 model.

baseline with CAL on the base data set using i.i.d. random splits, we observe that the IR-net baseline performs well on both the random splits and the covariate shift data sets. This is consistent with previous observations that covariate shifts are not particularly challenging for neural network models, as shown in Table 1. This performance indicates that neural networks tend to overfit to the training distribution. Consequently, the performance on OOD test sets is significantly worse than on i.i.d. data sets, highlighting the overfitting issue. In contrast, the CAL model demonstrates more consistent performance across OOD scenarios. While it shows a higher mean absolute error (MAE) of 0.5424 eV on the random split set, 0.4925 eV on the covariate shift set, and 0.6115 eV on the relation shift set, it maintains better performance stability across OOD data compared to the IR-net baseline. This stability reflects the effectiveness of the CAL approach in handling OOD data, even at the cost of slightly reduced performance on data similar to the training data set. This baseline model comparison was conducted with the same hyperparameters across all models to maintain fairness. However, based on the original Self-Adversarial Learning (SAL) paper, the CAL model theoretically has the potential to achieve better performance if the number of training iterations is significantly increased. The current results reflect a balance between training time and performance. It is interesting to note that our CAL iid performs best on the covariate shift set, even surpassing the performance on the i.i.d. random split set. This can be attributed to the combination of partial training and adversarial attacks. The covariate shift samples are selected by identifying outliers in the feature X domain. Although these samples are removed and unseen by the CAL model, they have neighbors in the feature map that remain in the training set. These samples exhibit high losses when processed through the network due to their deviation from the majority of the sample distribution. The partial sampling mechanism effectively selects these outliers as seed samples for adversarial attacks. Compared to the IR-net baseline, which shows clear overfitting under the same training hyperparameters, our CAL iid demonstrates improved performance on the prior shift set. This performance on covariate shift indicates the substantial potential of CAL for handling OOD scenarios. Our CAL method facilitates adversarial learning using a specific secondary training set rather than sampling from the original training set. This approach requires a base training set and a secondary, smaller set of *labeled* training data from a specific domain where we aim to improve model performance. The standard approach for this problem in the literature is fine-tuning: training a base model and then updating it with a new training set. We investigate whether our targeted CAL can outperform standard fine-tuning when we have a small number of labeled examples from the target distribution, in addition to a larger training set with potential distribution shifts. Table 2 shows the performance of these models at the bottom. For larger OOD covariate shift and prior shift data sets (with 1024 samples used for training), our model achieves modest performance improvements over the baseline, with 65.46 and 90.85% lower losses, respectively. Since the fine-tune and targeted CAL are trained toward the three OOD test set respectively, the performance of the i.i.d random split test set are the average of the 3 models. Our targeted CAL model have 37.55% performance improve while the fine-tune method suffered significant overfitting issue with 92.01% performance loss. Meanwhile, the targeted CAL outperforms the IR-net fine-tune by 21.3% on the prior shift test set, with a higher improvement ratio. Considering that our targeted CAL, unlike fine-tuning, does not require a baseline model to start training, achieving similar or even better results demonstrates the capability of our CAL model. For the piezoelectric data, with only 256 samples for training, we observe a much more significant improvement: a 51.12% lower loss for the targeted CAL, with an 84.75% improvement over the baseline, whereas the fine-tune method only improved by 14.17% over the baseline.

Additionally, we gained insights into our OOD data set by evaluating the performance of targeted versus fine-tuned results. Figure 1(b) indicates that the majority of prior shift samples are near the training set in terms of Y values, and Figure 2 shows the overlap between prior shift and training set samples. We believe this explains why the IR-net fine-tune model performs better on the prior shift than on the covariate shift. In contrast, the targeted CAL generates adversarial samples that guide the model toward better stability. It is also worth noting that with small data sets, the fine-tuning strategy can lead to severe overfitting, resulting in performance degradation compared to the baseline, as indicated by the relation shift test results in Table 1.

We also evaluated the performance of targeted CAL for bandgap prediction and compared it with the baseline CAL algorithm without targeting. The OOD test sets were selected from the 84,190 MP samples, as discussed in the Data set section. In this experiment, our prior shift set was selected using bandgap energy, and the performance results are shown in Table 3. We observe similar and even better performance advantages compared to the fine-tuning method, as shown previously. Not only does our targeted CAL outperform the fine-tuning method on all OOD shift sets, but it also shows better improvement over its baseline model. In terms of data distribution, we see a significant performance increase for both models on the prior shift set, with more than 60% improvement over the baseline for both models. As before, the fine-tuning method shows a minor improvement on the relation shift test set, with a 7.02% improvement rate,

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Table 3. Targeted CAL Model Demonstrates Better MAE OOD Performance in Bandgap Prediction (eV) across all OOD Shift Test Sets

	train	random splits	covariate shift	prior shift	relation shift
IR-net baseline	0.0259	0.3214	0.6178	2.8085	0.8176
CAL iid	0.4841	0.9242	1.0686	3.0002	1.0649
IR-net fine-tune		0.6487	0.4705	1.0276	0.7602
targeted CAL		0.9878	**0.3521**	**0.4425**	**0.2524**
% fine-tune improvement		−101.83%	34.04%	63.41%	7.02%
% CAL Improvement		−6.89%	67.05%	85.25%	76.29%

indicating overfitting on the visible data set. In contrast, our targeted CAL achieved a 66.79% better performance. This data set, demonstrating the largest improvement of our method for both bandgap and formation energy prediction and informed directly by existing crystal OOD challenges, merits further consideration.

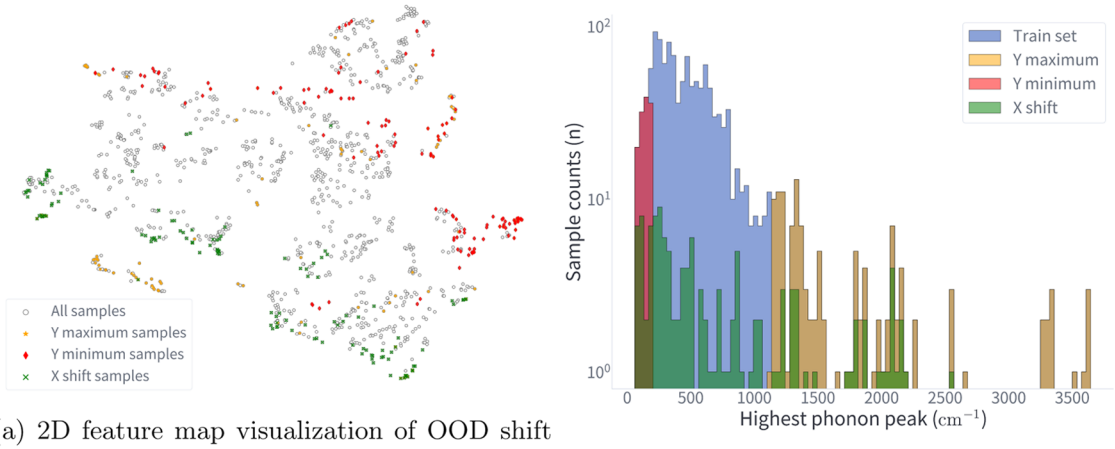
Piezoelectric Materials: A Case Study. Due to the small size of the data set (317 samples), we investigate whether varying the amount of samples held out for validation affects the results. Table 4 shows that fine-tuning fails to leverage additional data effectively, whereas our modified CAL continues to improve with an extra 128 training data points used to generate adversarial samples. Given the small data set, we hypothesize that CAL functions as a form of data augmentation, providing the model with more data to learn from. This newly generated adversarial data helps prevent overfitting and enhances learning. Future work could explore other data augmentation techniques for low-data regimes in materials science and evaluate our method's performance on additional small materials science data sets. Ultimately, any model will perform optimally on a specific subset of data, underscoring the importance of precise assessment when applying machine learning models for downstream material property prediction tasks. Training and validation set performance in these experiments do not strongly correlate with performance on specific types of downstream test data with distribution shift.

Table 4. Piezoelectric Prediction (MAE) Performance Comparison: Smoothed Mean Squared Error for Models Predicting Formation Energy (eV) on Piezoelectric Data, Comparing Targeted CAL with Fine-Tuning

method	CAL	fine-tune	CAL	fine-tune
#finetuning samples	128	128	256	256
loss (held-out piezoelectric)	0.0774	1.4488	**0.0443**	0.0877

CAL Performance over Data Sets with Limited Samples. To further test the capability of our CAL model, we selected data sets from Matbench[41] with limited samples available for different properties, such as last phonon peak, shear modulus, and refractive index. Here, we focus on CAL performance in predicting last phonon peak, as a representative feature of the phonon-related properties., given their limited sizes and challenging target value distribution with a median absolute deviation of 323.78 cm^{-1}. Results for other data sets are available in the Supporting Files, Tables S1 and S2. We tested the performance on the last phonon peak data set with 1265 samples, the refractive index data set with 4764 samples, and the shear stress data set with 10,987 samples. We used different guidelines to determine the OOD test sets for these small data sets. We selected samples with the top 10% and bottom 10% of target values as our prior shift set. Since these data sets are already subsets of materials with specific properties, there is no clear subset to select for relation shifts, and thus no relation shift sets were chosen.

Figure 5(a) shows a two-dimensional projection of the data using UMAP, with test sets distinguished by colors. This projection shows that the test samples are positioned on the edges of the primary clusters, consistent with their OOD nature. Their distinct appearance in the visualization is aided by color coding, which highlights the differences in their distribution relative to the main data set. Notably, the values with low y are more heterogeneous, suggesting that low peaks may not be as indicative of composition as high peaks. Figure 5(b) illustrates the frequency of the highest frequency optical phonon mode peak, in units of cm^{-1}, for this last phonon peak prediction subset. The majority of samples have their phonon peak distributed between 0 and 1500 cm^{-1}, while some



(a) 2D feature map visualization of OOD shift samples

(b) OOD test sets target value distribution

Figure 5. Visualization of OOD test set samples: (a) Two-dimensional compositional feature space representation, with colored markers indicating individual samples' relationships to local clusters and global structure. (b) Distribution of target values for the OOD test sets, showing the frequency of the highest optical phonon mode peak. The y-axis is on a log scale.

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J. Phys. Chem. C 2025, 129, 6372−6385

samples have peaks as high as 3643.74 cm^{-1}. With a median absolute deviation of 323.78 cm^{-1}, even the best model on the Matbench leaderboard (as of May 2024) only achieved a mean absolute error of 28.76 cm^{-1} under standard benchmark conditions. Our method of identifying the OOD shift set adds to the challenge. For this experimental setup, our training set contains 799 samples, our regular test set includes 130 samples, and each OOD shift test set comprises 126 samples.

As shown in Table 5, the IR-net baseline achieves a smooth L1 training loss of 13.77 cm^{-1} and a testing loss of 43.71 cm^{-1}.

Table 5. OOD (MAE) Performance Comparison: Models Predicting Last Phonon Peak (cm^{-1})[a]

	train set	test set	covariate shift	maximum prior shift	minimum prior shift
IR-net baseline	13.77	43.71	232.90	770.39	38.42
CAL iid	142.14	143.63	445.53	1231.59	49.62
IR-net fine-tune		79.10	219.96	523.35	23.47
our targeted CAL		121.28	**76.10**	**195.23**	**27.13**
fine-tune % improvement		−80.96%	5.55%	32.06%	38.91%
targeted CAL % improvement		15.56%	**82.91%**	**84.14%**	**45.32%**

[a]Smoothed mean squared error is used during training and reported for the train set. The test set and all OOD data sets are evaluated by mean MAE. This data set is challenging due to the limited samples (1265) and extreme target value distribution with a median absolute deviation of 323.78 cm^{-1}. Our Targeted CAL model achieves significantly better results compared to the IR-net Fine-tune model across all three OOD test sets even though its baseline performs poorly.

Compared to the benchmark leaderboard, the test loss falls into the 10th place (as of May 2024), which is reasonable. On the other hand, the CAL iid struggles, with a high training loss of 142.14 cm^{-1} and a test loss of 143.63 cm^{-1}. It is just behind the random forest on the leaderboard. This outcome can be attributed to two key factors: (1) the data distribution in the i.i.d. scenario, which favored conventional training schemes which favors IR-net methods like IR-net, allowing it to achieve strong results; and (2) the nature of the CAL pipeline under the i.i.d. training method (referred to as CAL iid). For the sake of fairness and computational limitations, CAL iid was trained using the same approach as any conventional machine learning algorithm, without leveraging its key optimizations such as adversarial attacks, partial sampling, and tailored training schedules. These design choices were aligned with the IR-net baseline to enable direct comparison but do not reflect the full potential of the CAL pipeline when optimized for OOD scenarios. As shown in Figure 5(b), most minimum prior shift data are fairly near the training set distribution, hence the best loss of 49.62 cm^{-1} in all test sets for the CAL iid model. We also observed earlier that there is an overlap of samples from two prior shift sets in Figure 5(a). This suggests a complex relationship between the selected composition features and phonon peaks. This means that the hard-to-learn samples in the training set may not be similar to the OOD test sets. Figure 6 compares the models predicted values with the actual values, the x-axis represents the actual target values, while the y-axis represents the predicted output values, both in logarithmic scale.

Due to the limited size of the test sets, we opted for a batch size of 32 samples and used two batches for fine-tuning or targeting. Our targeted CAL model achieves significantly better results compared to the IR-net Fine-tune model across all three OOD test sets, even though its baseline performs poorly. For the X shift set, standard neural network machine learning models tend to have good performance, as shown previously. For phonon vibration prediction, however, the IR-net fine-tune model struggles to improve, ending up only 5.55% better than the baseline, while our targeted CAL has a loss of 76.10 cm^{-1}, becoming 82.91% better than the CAL i.i.d. For the maximum prior shift set, both baseline models have comparably high losses. Our targeted CAL model not only outperforms the IR-net fine-tuning but also achieves an i.i.d. test set loss of 62.85 cm^{-1} after fine-tuning, compared with the IR-net fine-tune

(a) Covariate shift test set performance (b) OOD test sets target value distribution

Figure 6. Baseline model performance on different test set for last phonon peak (cm^{-1}) prediction: (a) Predicted versus actual phonon peak for the Covariate shift set. The x-axis represents the actual target values, while the y-axis represents the predicted output values, both in logarithmic scale. The performance of the IR-Net and CAL i.i.d. is compared, with blue and red points representing their respective predictions. The dashed black line indicates the perfect fit where the predicted values exactly match the actual target values. (b) Comparison of predicted versus actual values across combined prior shift sets and the i.i.d. test set. The performance of both IR-Net and CAL i.i.d. is shown for maximum and minimum prior shifts, as well as the i.i.d. test set. The dashed black line represents the perfect fit. The red and blue points represent i.i.d. test set predictions, while pink and cyan indicate maximum prior shifts, and purple and orange show minimum prior shifts.

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model's i.i.d. test loss of 134.52. This demonstrates the robustness of our model after fine-tuning. Finally, our model also outperforms the IR-net's strong baseline for the minimum prior shift loss, with a 45.32% improvement and a loss of 27.13 cm^{-1}. All results suggest that although unrepresentative samples may cause significant performance drops, the partial training and adversarial attack module in our model are highly effective in problems with limited sample sizes when used to target a specific subset of data, akin to fine-tuning. Figure 7

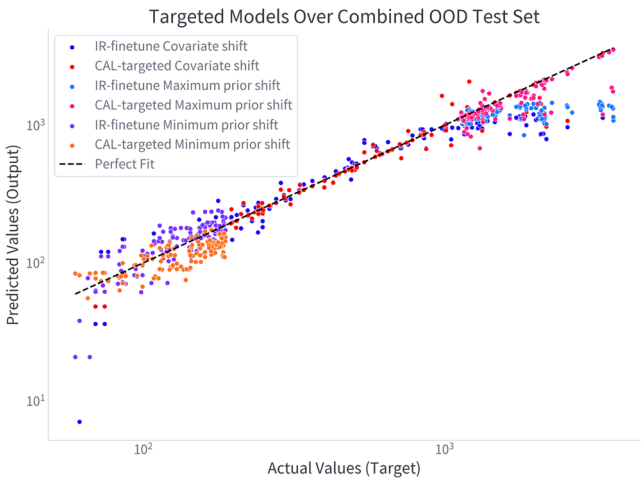

Figure 7. Comparison of predicted values versus actual last phonon peak (cm^{-1}) across multiple OOD test sets. The plot shows the performance of different models under various distribution shifts, including IR-finetune and CAL-targeted approaches for both covariate and prior shifts. The *x*-axis represents the actual target values, while the *y*-axis represents the predicted output values, both in logarithmic scale. The dashed black line represents the perfect fit, where predicted values match the actual target values exactly. The scatter points represent the model predictions under covariate shifts (blue and red) and prior shifts (minimum: purple, orange and maximum: pink, cyan). The variation in prediction accuracy highlights the differences in model performance, with CAL-targeted models demonstrating better alignment with the perfect fit line compared to the IR-finetune models, particularly in addressing covariate and maximum prior shifts.

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Ablation Study. Effects of the Partial Sampling and Adversarial Attack. To examine the effect of the proposed CAL pipeline, we compare the CAL, just Adversarial Attack (AA), just Partial Sampling (PS), and the baseline neural network (IRnet-3) as shown in Table 6. Compared to the baseline "IRnet-3" model, our CAL consistently achieves similar or lower loss values across all data conditions (Training Set, Random Splitting, Covariate Shift, Prior Shift, Relation Shift). This suggests that the combination of Adversarial Attacks (AA) and Partial Sampling (PS) within our approach effectively mitigates the negative effects of data distribution shifts. As mentioned earlier, neural network models have

Table 6. Smoothed MAE Performance of Our Baseline CAL Model and Its Ablated Versions[a]

method	training set	random splitting	covariate shift	prior shift	relation shift
our CAL	0.0061	0.0851	0.0856	0.1564	0.0781
AA	0.0163	0.0915	0.1067	0.1522	**0.0773**
PS	0.0081	0.0853	0.0900	**0.1403**	0.0801
IRnet-3	0.0059	0.0838	**0.0837**	0.1676	0.0823

[a]AA stands for the model with only adversarial attacks and PS for the model with only partial sampling. IRnet-3 is the barebone neural network model. We find partial sampling alone is less effective. Adversarial attack is effective when dealing with extremely small data sets.

certain stability against the covariate shift, while a simple adversarial attack brings more loss to the model if not well guided, hence the increase in loss. On the other hand, the Partial Sampling module alone shows some effectiveness against the covariate shift set. This is intuitively true since it is pushing the model to focus on high-loss samples where the *Y* values are in distribution and their *X* values may not be, adding an element of aleatory uncertainty. At the same time, the distribution of the formation energy in our prior set (Figure 1(b)) indicates that the majority of prior shift samples are near the training set in terms of *Y* value. The Partial Sampling technique can be beneficial for improving the performance of edge data from the training set, hence the leading result of the PS model for the Prior shift set. Finally, we notice that the AA module is highly efficient when dealing with extremely small data sets. The generated adversarial samples can be highly relevant to the target samples. The lack of data becomes advantageous. This finding prompted us to further improve our CAL pipeline by introducing the targeted training procedure.

One Layer Adversarial Gradient Attack. At the early stage of our experiment, we encountered a computational cost problem, where the gradient size increased explosively with the size of the data set and the complexity of the network. This is also the reason why we did not focus on more popular and complex structure-based crystal property prediction problems. To address the computational cost issue of our CAL model for complex neural network models, we proposed and evaluated the OOD performance for adjusted CAL models where gradients are derived from only one layer of the network. The Figure 8 shows the performance comparison of CAL models using gradients from different layers of the adversarial attack module, compared with the full-gradient CAL model. The backbone neural network of our CAL consists of three layers with a 2048−1024−512 structure. PG-2048 refers to the experiment where gradients for the adversarial attack are exclusively extracted from the 2048-unit layer, which demonstrates the best OOD performance among the tested configurations. In the in-distribution random splitting test set, we observed a gradual increase in loss as the gradient calculation moved from layer 2048 to layer 512, with our CAL performance falling between the 2048 and 1024 layers. The gradient calculation from the first layer, which corresponds to feature extraction,[35] yields the best performance among all layers. This pattern is consistent across the covariate shift and prior shift test sets. Additionally, the increased difficulty of the prior shift test set exacerbates the loss gap between different experiments. Finally, the relation shift test set, characterized by its complex relationships and

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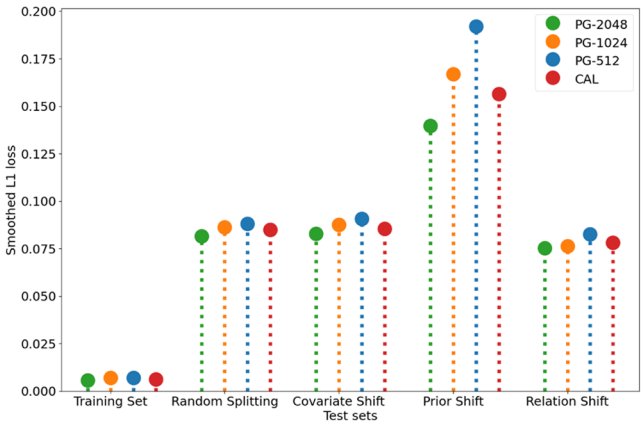

Figure 8. Comparison of smoothed Mean Absolute Error (MAE) performance for our baseline CAL model using partial gradients. The backbone neural network of our CAL consists of three layers with a 2048−1024−512 structure, and each color in the graph represents a partial gradient experiment focused on a specific layer. For instance, PG-2048 denotes the experiment where gradients for adversarial attacks are exclusively extracted from the 2048-unit layer.

limited samples, results in slightly worse performance for the CAL model compared to the 1024-layer.

■ DISCUSSION

In our CAL algorithm, the main goal of generating perturbed samples (compositions) is to identify training samples that are located in the area with high property variation. These perturbed samples themselves are NOT included in training the final model. Despite the fact that these perturbed samples may not even be able to be mapped to stable structures, in the composition space, they are mapped to the continuous area in the property space via our neural networks.

Limitations and Potential Weaknesses. While the Crystal Adversarial Learning (CAL) framework demonstrates its potential in addressing out-of-distribution (OOD) material property prediction challenges, it also comes with certain limitations that warrant further attention. One key limitation is the computational demand of the algorithm, particularly its significant GPU memory requirements. All experiments were conducted on a system equipped with an NVIDIA 4090 GPU (24 GB memory), utilizing PyTorch for implementation. The GPU memory limitations significantly influenced the choice of model architecture and parameter settings. Although more computational resources are available, the fact that a relatively small model like IR-net depletes the entire 4090 GPU is not acceptable. This concern becomes even more critical when considering future plans to deploy CAL over larger state-of-the-art models, where efficient resource utilization will be imperative.

Additionally, the CAL algorithm is tailored to scenarios where a small portion of OOD samples is available during training. While this design allows for simultaneous training with i.i.d. and OOD data, it may not fully address situations where OOD data is unavailable or severely imbalanced. The unique structure of CAL also relies heavily on IR-net as the base model due to its lightweight architecture, further narrowing its application scope until computational efficiency is improved.

The current implementation focuses on demonstrating the utility of the CAL framework rather than achieving comprehensive adaptability across various architectures. Furthermore, the reliance on adversarial learning introduces inherent sensitivity to parameter tuning, which could affect model stability and robustness if not optimized appropriately.

Future work will aim to address these limitations by exploring strategies to reduce computational demands, enabling broader compatibility with diverse architectures and data sets. Additionally, more thorough sensitivity analyses and benchmarking against alternative methods will be pursued to further validate and refine the CAL framework. These efforts will help ensure that CAL not only meets the current challenges of materials science OOD scenarios but also evolves into a more robust and widely applicable tool.

■ CONCLUSIONS

In materials property prediction, shifts in distribution are unavoidable and cause significant problems for standard machine learning models. Our investigation confirms that existing material property prediction models struggle with out-of-distribution (OOD) data. This limitation leads to unreliable predictions and hinders real-world applications in material science. To address this challenge, we explored the impact of various distributional shifts, including covariate shift (variations in input features), prior shift (variations in target labels), and relation shift (changes in the underlying relationship between inputs and targets). Our experiments showed that our CAL models generalize better overall, sometimes at the cost of performance on data similar to the training data. For small data sets such as the piezoelectric property prediction problem, CAL can adapt a base model more effectively than fine-tuning, the dominant paradigm in transfer learning. We find that no single model outperforms the others on all data sets. As such, it is vital that materials property models are carefully evaluated on data that is as close as possible to the data sets to which they will be applied. Our work makes three key contributions. First, by constructing custom data sets exhibiting three different distribution shifts, we demonstrate the severity of OOD issues in current ML models for property prediction. This finding highlights a critical limitation that needs to be addressed for robust material property prediction. Second, we propose a novel approach for OOD material property prediction that integrates adversarial learning with training set ranking, which effectively enhances model robustness under diverse test scenarios marked by different input and target distributions. Finally, in our experiments simulating realistic prediction scenarios, we demonstrate the usefulness of our adversarial learning method over the baseline training and fine-tuning approach for OOD property prediction, especially when training with limited data. These results demonstrate the CAL model's capability to handle data scarcity, a common challenge encountered in material science research. In conclusion, our research contributes to advancing OOD methods for materials science, providing a foundation for future exploration. It offers immediate practical utility by providing a robust solution for overcoming OOD limitations typical in extrapolative material property prediction. Additionally, it opens doors for further exploration in this crucial area, paving the way for more reliable and generalizable material property prediction models.

■ ASSOCIATED CONTENT

Data Availability Statement

All data used in this study were collected or generated from publicly available databases and packages. The generated data

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J. Phys. Chem. C 2025, 129, 6372−6385

set and the code for the CAL framework are accessible on GitHub at https://github.com/ramborz/CAL.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpcc.4c07481.

Smoothed mean squared error for models predicting the base-10 logarithm of the DFT Voigt−Reuss−Hill average shear moduli; smoothed mean squared error for models predicting the dielectric refractive index; CAL targeted toward the Prior shift hyper-parameter search on batch size; performance of CAL model predicting last phonon peakcm^{-1}across multiple independent training runs with different random seed initializations; box-plot illustrating the mean absolute error (MAE) distribution across different test sets and models, and UMAP embedding of the MP data set with density represented by the depth of color (PDF)

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Conceptualization, J.H.; Methodology, Q.L., J.H., N.M.; Software, N.M., Q.L.; Resources, J.H.; Writing—original draft preparation, Q.L., J.H., N.M.; Writing—review and editing, J.H., N.M., Q.L.; Visualization, J.H., Q.L., N.M.; Supervision, J.H.; Funding acquisition, J.H.

Notes

The authors declare no competing financial interest.

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J. Phys. Chem. C 2025, 129, 6372−6385